

STAT 5309 – SPRING 2018

LAB 0

**** Contents: R Introduction**

***Due: Feb 24**

A. PRACTICE

-----Use R as a calculator

```
>log(2, base=10)
>log(2)          #natural log
>exp(0.6)
```

##-----Reading data into R

#-----Reading small data files: c() and scan()

```
>data1 <- c(1,2,3, 4,5,6) ; data1    # a vector of numerics
>data2 <- c("Mon", "Tues", "Wed", "Thurs", "Fri", "Sat", "Sun")  #a vector of characters
>data2
```

```
>data3 <- scan()          #enter data as numeric, commas are not needed
>data4 <- scan(what = "character")    #enter text
>scan(sep =",")          #if data are numeric, separated by commas,
>scan(sep="," , what ="char") #if data are character , separated by commas.
```

#---Read from a directory

```
>setwd("C:/Users/USER/Documents/Data ")    #make default directory; use forward slashes.
>getwd()    #default directory
```

#-----Read Excel files: read.csv(), read.delim() (saved as .csv (comma separated version))

```
> data<- read.csv(file.choose(), sep=",", header=TRUE)
> data <- read.csv(file.choose(), sep=",", header=TRUE, row.names =1)
```

```
# -----Read text file: read.table() (file save as .txt)
>data<- read.table(file.choose(), header=TRUE)
>data
```

Note: Missing data are denoted as NA's (not available)

-----View objects in R: all items you stored in the R environment are called **objects**.

```
>ls()           #retrieve all items stored; all items should be saved with a name
>ls(pattern="Ad") #retrieve all which contain "Ad"
>rm(grades)      #remove "grades"
```

-----Objects in R: vectors, matrices, data frames of numeric data, character data;

---Vectors (column vectors); Matrices

```
>numbers <- c(1,2,3,4,5)
>class(numbers)
[1] "numeric"           # vector of numerics
> days <- c("Mon", "Tues", "Wed")
> days
[1] "Mon" "Tues" "Wed"      # vector of characters, in ""
> class(days)
```

#-----Convert numeric data to character data ;

```
>numbers_char <- as.character(numbers)
>numbers_char
[1] "1" "2" "3" "4" "5"
```

#-----Matrices: A matrix is a two-dimensional data object, *with rows and columns*. A matrix can be a single row or a single column.

```
> row1<- c(1,2,3,4)
> row2 <- c(3,4,5,6)
> row1
[1] 1 2 3 4          #a column vector
```

#Form a matrix using vectors: cbind() , rbind()

```
> mat1 <- cbind(row1, row2)      #cbind(): combine vectors as columns
> mat1
```

	row1	row2
[1,]	1	3
[2,]	2	4
[3,]	3	5
[4,]	4	6

```
> mat2 <- rbind(row1, row2)      #rbind(): combine vectors as rows
> mat2
```

	[,1]	[,2]	[,3]	[,4]
row1	1	2	3	4
row2	3	4	5	6

```
> class(mat1)
```

```
[1] "matrix"      #mat1 is a matrix object
> mat1[1,1]     #retrieve an entry
```

[1]	1
-----	---

```
> mat1[1, ]     #retrieve the 1st row
```

row1	row2
1	3

```
> mat1[, 1]     #retrieve 1st column
> mult <- mat2 %*% mat1  # multiply matrices using % * %
> mult
```

row1	row2
row1	30 50
row2	50 86

#-----Data frames: a data frame is a two-dimensional object,(which looks like the matrix object). All data frames are rectangular. R handles data file in table form.

(To perform matrix operations (addition/subtraction/multiplication/transpose/inverse), we require the data frames converted into matrix object)

```
> class(airfares)
[1] "data.frame"      #airfares is a data frame
```

```
> airfares
```

	City	Fare	Distance
1	1	360	1463
2	2	360	1448
3	3	207	681
4	4	111	270
5	5	93	190
....			

```
> dim(airfares)      #dimension of the data frame
[1] 17 3
```

```
> colnames(airfares)
[1] "City" "Fare" "Distance"
> rownames(airfares)
[1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11" "12" "13" "14" "15" "16" "17"
```

```
> airfares$Fare #retrieve the 2nd column of data frame
[1] 360 360 207 111 93 141 291 183 309 300 90 162 477 84 231 54 429
```

```
#-----attach(), detach()-----
```

```
> attach(airfares) # attach the data frame to the R environment
> Fare # $ symbol is not needed
[1] 360 360 207 111 93 141 291 183 309 300 90 162 477 84 231 54 429
> detach(airfares) #detach the data frame from the R environment
```

```
> airfares_mat <- as.matrix(airfares) #convert data frame airfares as a matrix
> airfares <- data.frame(airfares_mat) #convert back to data frame
```

```
#-----Combine vectors into data frames:
```

```
> days<- c("Mon", "Tues", "Wed", "Thurs") #days is a vector
> sales <- c(2,3,4,1)
> mysales <- data.frame(rbind(days, sales))
> mysales
```

	X1	X2	X3	X4
days	Mon	Tues	Wed	Thurs
sales	2	3	4	1

```
# R assigns column (variable) names: X1, X2, X3, X4
```

```
> mysales_2 <- data.frame(cbind(days, sales))
> mysales_2
```

	days	sales
1	Mon	2
2	Tues	3
3	Wed	4
4	Thursday	1

```
# R not assign column names , since column names already exist
```

##-----Saving your work in R-----

#-----**Save Workspace:** R saves all the objects currently into memory to the default workspace file in the default working directory. Extension .RData

>save.image("C:\\Users\\USER\\Desktop\\STAT5309\\Week1.RData") #save.image(), or
use the Save Workspace from the File Menu. File name: Week1.RData

#-----Save R objects in a directory

Suppose we have an object g, which is a R linear model object

> save(g, file="C:/Users/USER/Desktop/STAT4310/airfares.RData")
#g will be saved in the file airfares.RData

#-----Load data files or objects in to R.

>load("C:\\Users\\USER\\Desktop\\STAT4310\\Week1.RData") #Use load() or double
click the file in the directory

##-----Miscellaneous

> summary(airfares) #Summary of basic statistics of the variables in airfares

#-----Factors

>height <- c(132,151,162,139,166,147,122)
>weight <- c(48,49,66,53,67,52,40)
>gender <- c("male","male","female","female","male","female","male")
>data <- data.frame(cbind(height, weight, gender))

	height	weight	gender
1	132	48	male
2	151	49	male
3	162	66	female
4	139	53	female
5	166	67	male
6	147	52	female
7	122	40	male

>is.factor(data\$gender) # Test if the gender column is a factor.

[1] TRUE

>data\$gender # Print the gender column so see the levels.

[1] male male female female male female male

Levels: female male

B. EXERCISE

1. Distributions

- Histogram of normal distributed sample data

```
s1 <- rnorm(100, 0,1)
```

```
hist(s1)
```

- Density curve of normal distribution

```
x <- seq(-4, 4, .01)
```

```
hx <- dnorm(x, 0,1)
```

```
plot(x=x, y=hx, pch=16, cex= .01)
```

- Density curve of some t-distribution

```
t <- seq(-4,4, .01)
```

```
ht <- dt(t,10)
```

```
plot(t, ht, pch=16, cex=.01)
```

- Density curve of some chi-squared distribution

```
chi_sq <- seq(0, 50,.5)
```

```
hchi_sq <- dchisq(chi_sq, 5)
```

```
plot(chi_sq, hchi_sq, pch=16, cex=.01)
```

- Density curve of some F-distribution

```
f <- seq(0, 50, .5)
```

```
hf <- df(f, 3, 20)
```

```
plot(f, hf, pch=16, cex=.1)
```

Save all plots, reduced.

2. Check Normality Distribution of data

```
data <-c(26.4,23.5,25.4,22.9,25.2,39.2,25.5,31.9,26.0,44.6,35.5,38.6,  
30.1,31.0,30.8,32.8,47.7,39.1,55.3,50.7,73.8,71.1,68.4,77.1,  
19.4,19.3,18.7,19.0,23.2,21.3,23.2,19.9,18.9,19.8,19.6,21.9)
```

```
hist(data)
```

```
data_s <-sort(data)           #sort data increasing
```

```
rank <- rank(data_s)         #rank data from 1 to 36
```

```
size <- length(data_s)       # size of data
```

```
p <- (rank-.5)/size          #cummulative prob of data
```

```
z_quantile <- qnorm(p)       # Standard Normal quantiles with such probability
```

```
plot(x=z_quantile, y=data_s, pch=16) #scatterplot of x=Z quantiles, y= data sorted
```